

Management of Locally Advanced NPC

Department of Radiation Oncology
Taichung Veterans General Hospital

Yi-Chun Liu

Outline

1. Treatment failure of current Tx
2. Optimal chemoradiotherapy
3. Target therapy in NPC

NPC Treatment

Radiotherapy

±

Chemotherapy

Patterns of failure treated by non-IMRT technique

Series	Cases	Treatment	Follow-up (median)	Failure site(s)			
				Local	Regional	Distant	Locoregional
Hong et al. [32]; SYSU, China	411	RT	NA	76 (18.5 %)	20 (4.9 %)	82 (20.0 %)	
Ma et al. [2]; SYSU, China	621	RT	60 m	109 (17.6 %)	44 (7.1 %)	136 (21.9 %)	
Yeh et al. [3]; KSCGMH, Taiwan	849	RT	112 m	188 (22.1 %)	55 (6.5 %)	227 (26.7 %)	
Lee et al. [7]; HK	2,687	RT±Ch	40.8 m	12 %	5 %	16 %	
Leung et al. [6]; TMH, HK	1,070	RT±Ch	56 m	190 (17.8 %)	72 (6.7 %)	245 (22.9 %)	
Lu et al. [33]; JSCH, China	159	RT	61 m	18 (11.3 %)	12 (7.5 %)	33 (20.8 %)	
Liu et al. [34]; SYSU, China	749	RT±Ch	62 m	110 (14.7 %)	32 (4.3 %)	133 (17.8 %)	
Lai et al. [31]; SYSU, China	764	RT±Ch	52.8 m	87 (11.4 %)	34 (4.5 %)	134 (17.5 %)	
Sum of above	7,319			1100 (15.0 %)	403(5.5 %)	1666 (22.8 %)	
Cheng et al. [35]; KFSYSCC, Taiwan	630	RT±Ch	58 m	NA	NA	96 (15.2 %)	60 (9.5 %)
Yi et al. [36]; CHCAMS, China	905	RT	105 m	NA	NA	150 (16.6 %)	130 (14.4 %)

IMRT intensity-modulated radiotherapy, RT radiotherapy, NA not available, Ch chemotherapy, m month, SYSU Sun Yat-sen University, KSCGMH Kaohsiung Chang Gung Memorial Hospital, HK Hong Kong, TMH Tuen Mun Hospital, JSCH Jiangsu Cancer Hospital, KFSYSCC Koo Foundation Sun Yat-Sen Cancer Center, CHCAMS Cancer Hospital of Chinese Academy of Medical Sciences

Patterns of failure treated by **IMRT** technique

Series	Cases	Stage	Treatment	Follow-up (median)	Failure site(s)		
					Local	Regional	Distant
Kwong et al. [37]; QMH, HK	50	III-IVB	RT±Ch	25 m	4 (8.0 %)	2 (4.0 %)	2 (4.0 %)
Wolden et al. [38]; MSKCC, US	74	I-IVB	RT±Ch	35 m	6 (22.1 %)	NA	NA
Kam et al. [39]; PWH, HK	63	I-IVB	RT±Ch	29 m	4 (6.3 %)	1 (1.6 %)	13 (20.6 %)
Lee et al. [40]; UCSF, US	67	I-IVB	RT±Ch	31 m	1 (1.5 %)	1 (1.5 %)	17 (25.4 %)
Xiao et al. [41]; SYSU, China	81	III-IVA	RT+Ch	54 m	4 (4.9 %)	2 (2.5 %)	14 (17.3 %)
Lee et al. [42]; RTOG0225	68	I-IVB	RT±Ch	31.2 m	5 (7.4 %)	6 (8.8 %)	10 (14.7 %)
Lin et al. [43]; CHFJMU, China	323	II-IVB	RT±Ch	30 m	12 (3.7 %)	6 (1.9 %)	26 (8.0 %)
Tham et al. [44]; Singapore	195	I-IVB	RT±Ch	36.5 m	19 (9.7 %)	5 (2.6 %)	19 (9.7 %)
Wong et al. [45]; TMH and QMH, HK	175	I-IVB	RT±Ch	34 m	10 (5.7 %)	11 (6.3 %)	20 (11.4 %)
Ng et al. [46]; PYNEH HK	193	I-IVB	RT±Ch	30 m	16 (8.3 %)	9 (4.7 %)	23 (11.9 %)
Lai et al. [31]; SYSU, China	512	I-IVB	RT±Ch	52.8m	29 (5.7 %)	14 (2.7 %)	72 (14.1 %)
Bakst et al. [47]; MSKCC, US	25	II-IVB	RT+Ch	33 m	2 (8 %)	2 (8 %)	2 (8 %)
Sum of above	1,826				112 (6.1 %)	59 (3.4 %)	218 (12.4 %)

Treatment Failure

1. Distant
2. Primary
3. Regional

Chemotherapy

Neoadjuvant?

Concurrent?

Adjuvant?

All meta-analyses favor CCRT

1. Huncharek et al. (Am J Clin Oncol 2002;25:219-23)
2. Langendijk et al. (J Clin Oncol 2004;15:4604-12)
3. Baujat et al. (Int J Radiat Oncol Biol Phys 2006;64:47-56)
4. Zhang et al. (BMC Cancer 2010;10:558)
5. Blanchard et al. (2014 ASCO abs. 6022) 4789 pt

Is CCRT the best ???

What is the limitation of meta-analysis?

Neoadj CT + RT vs. RT alone

	Neoadj CT + RT (N=40)	RT alone (N=40)	P-value
5-y OS	60%	48%	> 0.05
5-y DFS	55%	43%	> 0.05
5-y LR DFS	65%	68%	> 0.05
5-y MFS	74%	56%	> 0.05

(P 80 d1 + F 800 d2-5) q3w x 2 for 1988 AJCC I-IV

Median FU: 49 months; Patients received CT x 1, 2 = 5, 35

(Sapporo Medical University, Japan: Cancer 2002;94:2217-2223)

RT vs. RT + adj CT

	RT (n=116)	RT + VAC (n=113)	P-value
--	---------------	---------------------	---------

RFS	55.8%	57.7%	0.45
-----	-------	-------	------

OS	67.3%	58.5%	0.13
----	-------	-------	------

Patterns of failure are similar

CT: VCR 1.2 d1, CTX 200 po d1-4, ADR 40 d1, q4w x 6;

1978 UICC stage II-IV; median FU: 43 months

(Istituto Nazionale Tumori, Milan, Italy: J Clin Oncol 1998;6:1401-1410)

CCRT?

Or

CCRT + Adj CT?

CCRT + Adj CT vs. RT alone

	CCRT + Adj CT (N=78)	RT alone (N=69)	P-value
3-y PFS (5-y)	69% (58%)	24% (29%)	<0.001
3-y OS (5-y)	78% (67%)	47% (37%)	0.005
Toxicity	↑		

1988 AJCC/UICC Stage III-IV (n=193)

CCT: P 100 d1, 22, 43; AdjCT: (P80 d1, F 1000 d1-4, q3-4w) x 3

Compliance: CCT (63%), AdjCT (55%)

(RTOG, SWOG, ECOG: J Clin Oncol 1998;16:1310-1317, Proc ASCO 2001 update)

CCRT + Adj CT vs. RT alone

Table 4 Randomized trials of CCRT+AdjCT vs RT alone

Series	Stage	Cases	*Concurrent CT; AdjCT	Follow-up (median)	5-y OS CCRT+AdjCT vs RT	5-y PFS/DFS/FFS CCRT+AdjCT vs RT
Al-Sarraf et al. [64, 81]; Intergroup 0099	II-IV	147	P100 q3w×3; P80F1000d1-4 q4w×3	>5 y	67 % vs 37 % (P=0.001)	58 % vs 29 % (P<0.001)
Wee et al. [73, 82]; Singapore	II-IV	221	P25d1-4 q3w x 3; (P20F1000) d1-4 q4w×3	4.9 y	65 % vs 46 % (P=0.0106)	55 % vs 45 % (P=0.0393)
Chen et al. [76]; SYSU, China	III-IV	316	P40 qw×7; P80F800d1-5 q4w×3	29 m	89.8 % vs 79.7 % (P=0.003)	84.6 % vs 72.5 % (P=0.001)
Lee et al. [72, 83]; NPC 9901, HK	III-IV	348	P100 q3w×3; P80F1000d1-4 q4w×3	2.3 y		
Lee et al. [75, 84]; NPC 9902, HK	III-IV	189	P100 q3w×3; P80F1000d1-4 q4w×3	2.9 y		
Lee et al. [85]; NPC 9901+NPC 9902	III-IV	441	P100 q3w×3; P80F1000d1-4 q4w×3	6.1 y	72 % vs 63 % (P=0.037)	68 % vs 55 % (P=0.002)

Compliance of ConCT + AdjCT

Trial	ConCT	AdjCT
Intergroup 0099	63%	55%
RTOG 0225	88%	46%
RTOG 0615	68.2%	47.7%
(Avastin	69.8%	48.8%)
Singapore	71%	54%
Hong Kong 9901	52%	76%
Hong Kong 9902	23-53%	63-73%
China SYSU	68%	61%
China SYSU 2012	41%	-
	45%	63%

CCRT vs. RT alone

Series	Stage	Cases	Concurrent chemotherapy	Follow-up (median)	5-y OS CCRT vs RT	5-y PFS/RFS CCRT vs RT
Lin et al. [70] TCVGH, Taiwan	II-IV	284	(P20+F400)d1-4 q4w×2	65 m	72.3 % vs 54.2 % (P=0.0022)	71.6 % vs 55.0 % (P=0.0012)
Chan et al. [67, 88] PWH, HK	II-IV	350	P40 qw×8	66 m	70.3 % vs 58.6 % (P=0.065)	60.2 % vs 52.1 % (P=0.16)
Zhang et al. [74]; SYSU, China	III-IV	115	O 70 qw×6	24 m	100 % vs 77 % (P=0.01)	96 % vs 83 % (P=0.02)

CCRT (weekly P40 x 4-8) VS. RT (n=174+176=350), ↑ Toxicities
 Prince Wales & Queen Elizabeth Hospitals, Hong Kong

2002 JCO 2-y PFS = 76% vs. 69% (P=0.10)

2005 JNCI 5-y PFS = 60.2% vs. 52.1% (P=0.16)

5-y OS = 70.3% vs. 58.6% (P=0.065)

SYSU 115 p't, AJCC 1997 stage III/IV NPC median F/U: 114 months
 (weekly oxaliplatin)

5-y OS=73.2% vs. 60.2% (P=0.028)

Ann Oncol 2013;24:2131-6

5-y DMFFS=74.7% vs. 63.0% (P=0.027)

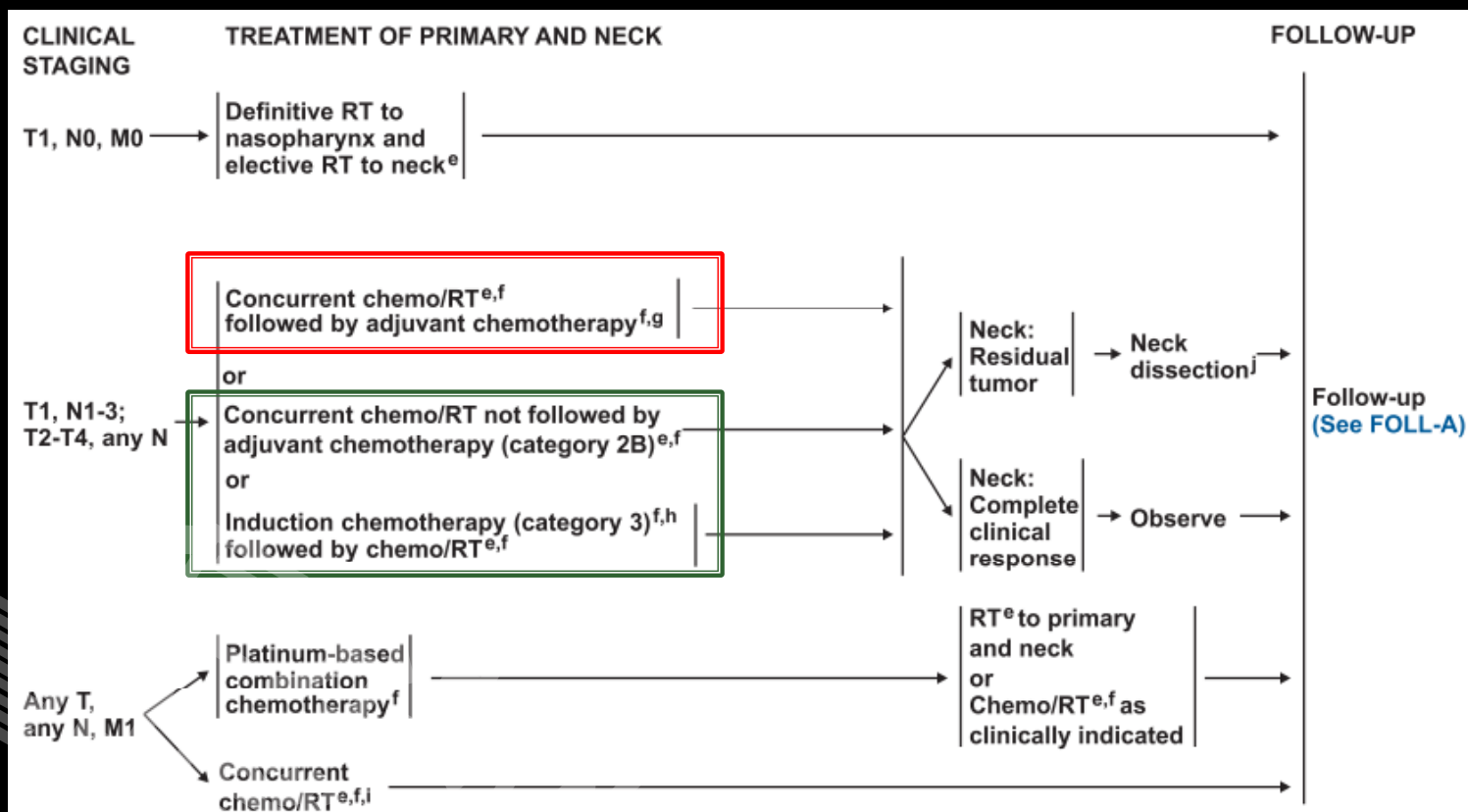
NCCN Guideline?

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Head and Neck Cancers

Version 1.2014

NCCN.org



Nasopharynx:

- Chemoradiation followed by adjuvant chemotherapy
 - Cisplatin + RT followed by cisplatin/5-FU^{18,19} or carboplatin/5-FU²⁰ (category 2B for carboplatin/5-FU) Tailand, n=50, Carbo-Adj Carbo/5FU
 - Cisplatin + RT without adjuvant chemotherapy²¹?

¹⁸JCO 1998;16:1310-1317

¹⁹JNCI 2005;97:536-539 CCRT only

²⁰Head Neck Oncol 2011;3:30

²¹Lancet Oncol 2012;13:163-171

²³NEJM 2007;357:1705-1715 (TAX324)

²⁶EJC 2007;43:1399-1406

²⁸Cancer Chemo Pharmacol 2010;65:589-595

²⁹JCO 2009;27:242-249

Phase I/II, n=45,
3-y OS 56%, PFS 56%

Nasopharynx:

- Induction (category 3)/Sequential chemotherapy
 - Docetaxel/cisplatin/5-FU²⁸
 - Docetaxel/cisplatin (category 2B)²⁹
 - Cisplatin/5-FU²³
 - Cisplatin/epirubicin/paclitaxel?
 - Following induction, agents to be used with concurrent chemoradiation typically include weekly cisplatin¹⁹? or carboplatin²⁶?

Phase II study of docetaxel, cisplatin, and 5-FU induction chemotherapy followed by chemoradiotherapy in locoregionally advanced nasopharyngeal cancer

Woo Kyun Bae · Jun Eul Hwang · Hyun Jeong Shim · Sang Hee Cho · Joon Kyoo Lee · Sang-Chul Lim · Woong-Ki Chung · Ik-Joo Chung Cancer Chemother Pharmacol (2010) 65:589–595
DOI 10.1007/s00280-009-1152-0

Abstract

Purpose This study sought to determine the feasibility and safety of induction chemotherapy with docetaxel, cisplatin, and 5-fluorouracil (5-FU) triple combination chemotherapy (TPF) followed by concurrent chemoradiotherapy (CCRT) for locoregionally advanced nasopharyngeal cancer (NPC).

Methods Patients with advanced NPC were treated with three cycles of induction chemotherapy. Docetaxel (70 mg/m^2) and cisplatin (75 mg/m^2) were given on day 1, followed by 5-FU ($1,000 \text{ mg/m}^2$) as a continuous infusion for 4 days. After induction chemotherapy, cisplatin was given at a dose of 100 mg/m^2 every 3 weeks with radiotherapy.

Results Thirty-three patients were enrolled; all patients were stage III ($n = 4$, 12.1%) or IV ($n = 29$, 87.9%). Among the patients, 32 patients completed both induction TPF therapy and CCRT, with responses as follows: five patients (15.2%) achieved a complete response (CR), and 27 patients (81.8%) a partial response (PR). At 6 weeks after CCRT, 23 patients (69.7%) had a CR and 9 patients (27.3%) a PR. The 3-year progression-free survival was 75.6% and the 3-year overall survival was 86.1%. Neutropenia (72.7%), febrile neutropenia (9.1%), and nausea (9.1%) were the most severe toxicities (grade 3–4) during induction chemotherapy, and mucositis (39.4%), fatigue (15.2%), and nausea (9.1%) were the most common toxicities (grade 3–4) during CCRT.

The role of
induction chemotherapy
in NPC

IndCT + RT vs. RT

IndCT + CCRT vs. CCRT

Table 6 Randomized trials of NeoCT+RT vs RT alone or NeoCT+CCRT vs CCRT

Series	Stage	Cases	NeoCT; Concurrent CT	Follow-up (median)	OS	PFS/DFS/FFS
Cvitkovic et al. [64]; INCSG	1987 N2-3	339	P100d1 E 70 d1 B15mg B12d1-5 q3w×3	49 m	3-y 60 % vs 54 % (P>0.05)	3-y 52 % vs 32 % (P<0.01)
Chua et al. [65]; AOCOA	Ho's T3, N2-3 N >3 cm	334	P60 E110 q3w×2-3	30 m		
Ma et al. [2]; SYSU, China	Chinese III-IV	556	P100d1 B10d-15 F800d1-5 q3w×2-3	62 m		
Pooled above 2 trials [87]	1997 AJCC II-IV	784		67 m	5-y 61.9 % vs 53.1 % (P=0.092)	5-y 50.9 % vs 42.7 % (P=0.014)
Hui et al. [77]; PWH HK	III-IV	65	P75 D75 q3w×2; P40 qw×8	4.3 y	3-y 94.1 % vs 67.7 % (P=0.012)	3-y 88.2 % vs 59.5 % (P=0.12)

NeoCT neoadjuvant chemotherapy, *RT* radiotherapy, *CCRT* concurrent chemoradiotherapy, *OS* overall survival, *PFS* progression-free survival, *DFS* disease-free survival, *FFS* failure-free survival, *y* year, *m* month, *INCSG* International Nasopharynx Cancer Study Group, *AOCOA* Asian-Oceanian Clinical Oncology Association, *SYSU* Sun Yat-sen University, *PWH* Prince of Wales Hospital, *HK* Hong Kong, *P* cisplatin, *E* epirubicin, *B* bleomycin, *F* fluorouracil, *D* docetaxel, *q3w* every 3 weeks, *qw* every week, *d* day

Summaries of recent IndCT vs. CCRT

Trial	Grouping	Endpoints
<u>HKNPCSG 0501</u> stage III/IV n=803	CCRT* + AdjCT (PFx3) PFx3 + CCRT* PXeloda + CCRT*	PFS OS (-)
<u>China</u> T3-4N1, T4, N2-3 n=480	TPFx3 + CCRT* CCRT*	FFS
<u>Singapore</u> stage III/IV n=172	GCarPax3 + CCRT** CCRT**	OS (-)
<u>TCOG</u> stage IV n=480	MEPFLx3 + CCRT** CCRT**	DFS
<u>TC-VGH</u> stage III/IV n=256	P-FLx10wk + RT alone CCRT***	OS (-) Less toxic

Target Therapy in Locally Advanced NPC

Literature Review

1. Hong Kong

(Ann Oncol 2012;23:1287-92)

2. Guangzhou, ENCORE study

3. Shanghai Fu-Da University

(J Cancer Res Clin Oncol 2013;139:1063-71)

4. RTOG-0615

Cetuximab

Bevacizumab

A phase II study of concurrent cetuximab–cisplatin and intensity-modulated radiotherapy in locoregionally advanced nasopharyngeal carcinoma

B. B. Y. Ma^{1*}, M. K. M. Kam¹, S. F. Leung¹, E. P. Hui¹, A. D. King², S. L. Chan¹, F. Mo¹, H. Loong¹, B. K. H. Yu¹, A. Ahuja² & A. T. C. Chan¹ (Ann Oncol 2012;23:1287-92)

¹State Key Laboratory in Oncology in South China, Sir YK Pao Centre for Cancer, Department of Clinical Oncology, Hong Kong Cancer Institute, The Chinese University of Hong Kong; ²Imaging and Interventional Radiology, Prince of Wales Hospital, Chinese University of Hong Kong, Hong Kong, China

30 LA NPC
Stage
III/IV=20/10

Erbixux 400/250 mg/m² weekly, 7-8 wks
Cisplatin 30 mg/m², starting d7-10, weekly, 6-7 wks
IMRT 74 Gy, starting d7-10, weekly, 6-7 wks

Endpoints:

1. Safety and feasibility of adding cetuximab to RT and weekly cisplatin
2. Loco-regional control rate at 3 months

Response rate:

CR: 83% } **96%**
PR: 13%

At Year 2:

- Progression-free survival rate: 86.5%
- Overall survival rate: **89.8%** (mOS not yet reached)
- Distant failure-free rate: 92.8%
- Locoregional failure-free rate: 93%

Tolerance

P30 range 3-7 wk, median=5
 ≥ 6 (50%), ≥ 5 (86%)
Interrupted in **60%** (Gr 3 mucositis)

E250 range 1-7 wk, median=5
 ≥ 6 (73%), ≥ 5 (93%)

Toxicities

Mucositis

Gr 3-4 mucositis 87% (26/30)

33% (10/30) on NG 3-200 d, median 41 d

Skin reaction

Gr 3 RT-dermatitis 20% (6/30)

Gr 3 Erbitux-rash 10% (3/30)

Hematology toxicity

Gr 3-4 lymphopenia 78% (24/30)

Gr 3 Leucopenia 10% (3/30)

ENCORE Trial

SYSUCC, Guangzhou etc. 8 hospitals

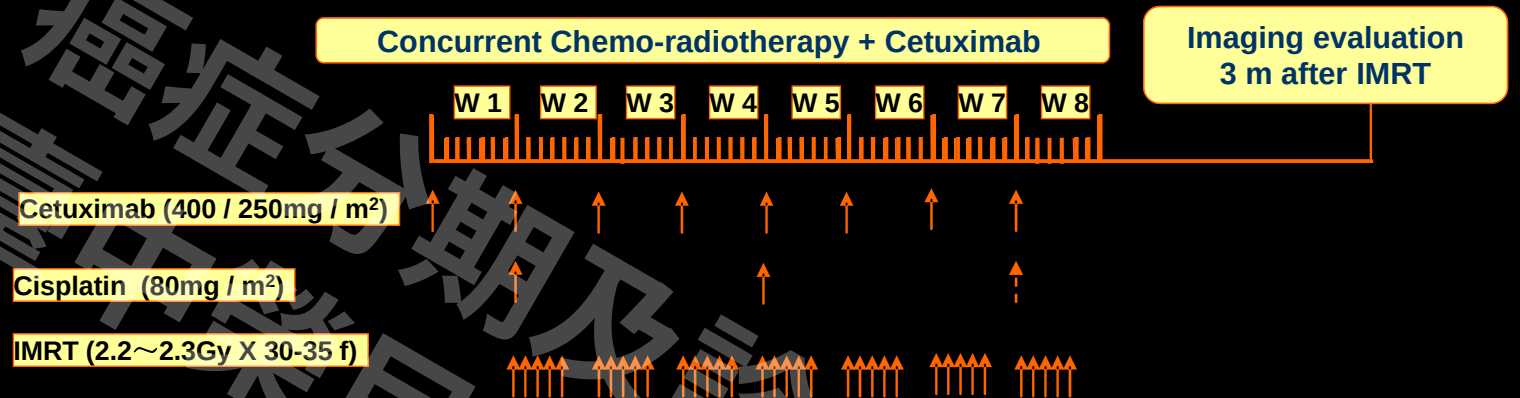
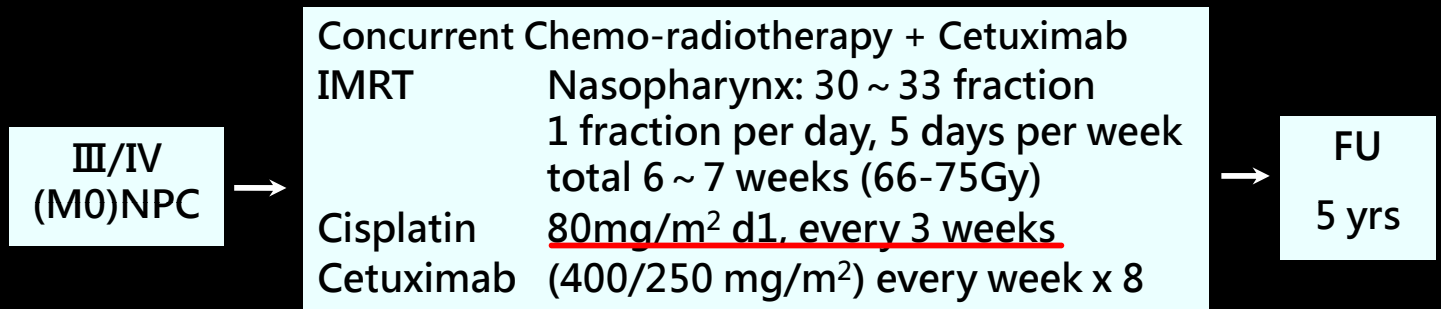
2008/07 – 2009/04

100 new NPC

stage III/IVa/IVb = 71%/22%/7%

Cancer Institute and Hospital, Chinese Academy of Medical Sciences
Sichuan Cancer Hospital and Research Institute
Fujian Cancer Hospital
Fudan University
Cancer Hospital of Guizhou Province
Guangxi Medical University
Central South University

ENCORE Study design

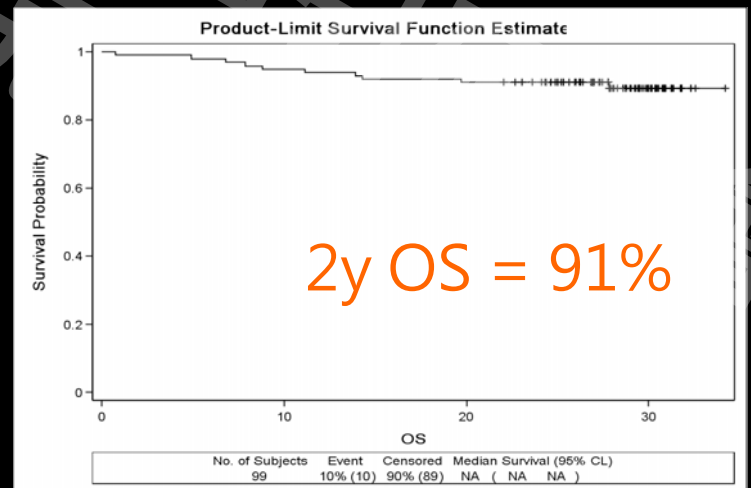
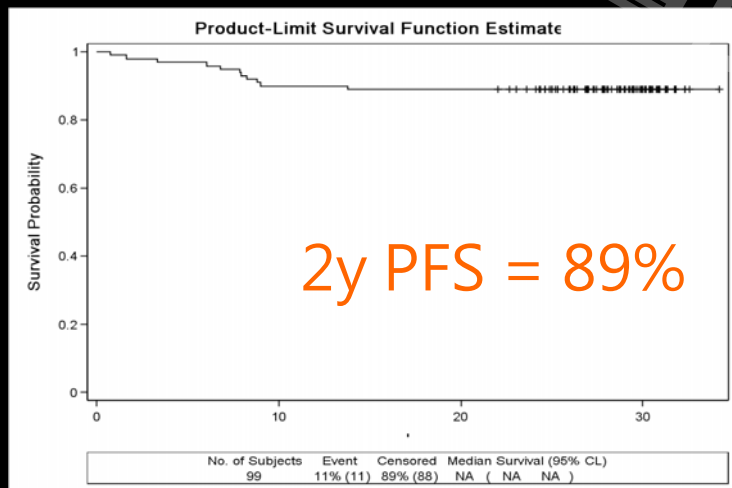


Compliance of C/T and Cetuximab

Cisplatin (cycle)	No. of cases	
1	1	1%
2	66	66%
3	33	33%
Complete as plan	99	99%
Cetuximab (week dose)		
2	1	1%
≥ 6	99	99%
Complete as plan	99	99%

Acute AEs

- Allergy 1 allergy to Cetuximab
 - Skin reaction
 - 64% $\geq$ Gr 2 skin rash since 3rd week
 - 58% Gr 1 dermatitis since the 3rd week
 - Mucositis
 - 90% $\geq$ Gr 2 mucositis since the 3rd week
 - 49% Gr 3 and 2% Gr 4 mucositis
 - Others
 - 40% $\geq$ Gr 2 xerostomia
 - 8%, 5%, 4% had $\geq$ Gr 2 ANC, Hb, Platelet
- ****One died of dysphagia 3 m after Tx



2012 ASCO

- Median FU = 23.5 months
- 2-y Local failure-free survival = 90%
- 2-y Neck failure-free survival = 90%
- 2-y Distant failure-free survival = 89%

Experience with combination of cetuximab plus intensity-modulated radiotherapy with or without chemotherapy for locoregionally advanced nasopharyngeal carcinoma

FDUCC, Shanghai
(J Cancer Res Clin Oncol 2013;139:1063-71)

Xiaoshuang Niu · Chaosu Hu · Lin Kong

2007-6 to 2010-12, 33 stage II/III/IV = 4(12%) / 11(33%) / 18(55%)

IMRT 66 Gy/30fr to T1-2; 70-70.4 Gy/32-35fr to T3-4

Cetuximab 400 → 250 mg/m²

Cisplatin-based C/T

Table 2 The type of therapy in all patients

The type of therapy	No. of patients
RT	3
Neo + RT	4
CCRT	12
Neo + CCRT	8
Neo + RT + Adj	5
Neo + CCRT + Adj	1

Toxicity

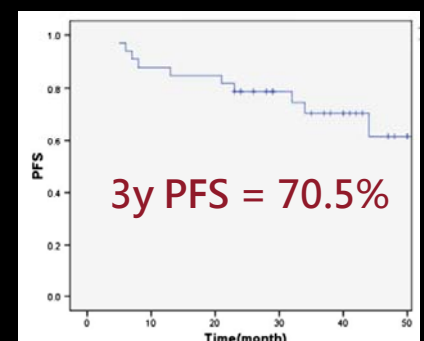
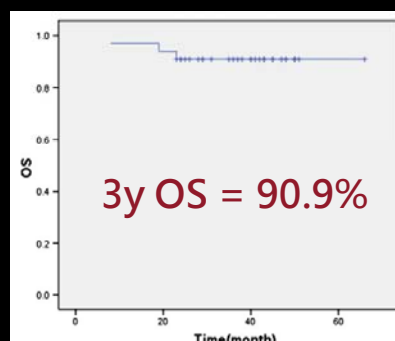
Gr 3/4 mucositis = 23 (69.7%) / 5 (15.2%) = 84.8%

Gr 2/3 RT dermatitis = 24 (72.7%) / 6 (18.2%)

Gr 2/3 skin rash = 18(54.5%) / 13(39.4%)

No Gr 3/4 hematological, renal & liver dysfunction

3-y DMFFS = 83.6%



(J Cancer Res Clin Oncol 2013;139:1063-71)

Addition of bevacizumab to standard chemoradiation for locoregionally advanced nasopharyngeal carcinoma (RTOG 0615): a phase 2 multi-institutional trial

Nancy Y Lee, Qiang Zhang, David G Pfister, John Kim, Adam S Garden, James Mechalakos, Kenneth Hu, Quynh T Le, A Dimitrios Colevas, Bonnie S Glisson, Anthony T C Chan, K Kian Ang

2006/12/13 to 2009/02/05

46 new NPC (stage IIb/III/IV=5/24/15)

IMRT 70Gy + P100 q3w x 3 + **Avastin 15 mg/kg q3w x 3**

+ AdjCT [(P80d1+F1000d-14) q4w + **Avastin 15 mg/kg q4w**] x 3

Median FU=2.5 yr

2-yr LRFSS=83.7%, 2-yr DMFFS=90.8%,

2-yr OS=90.9%, 2-yr PFS=74.7%

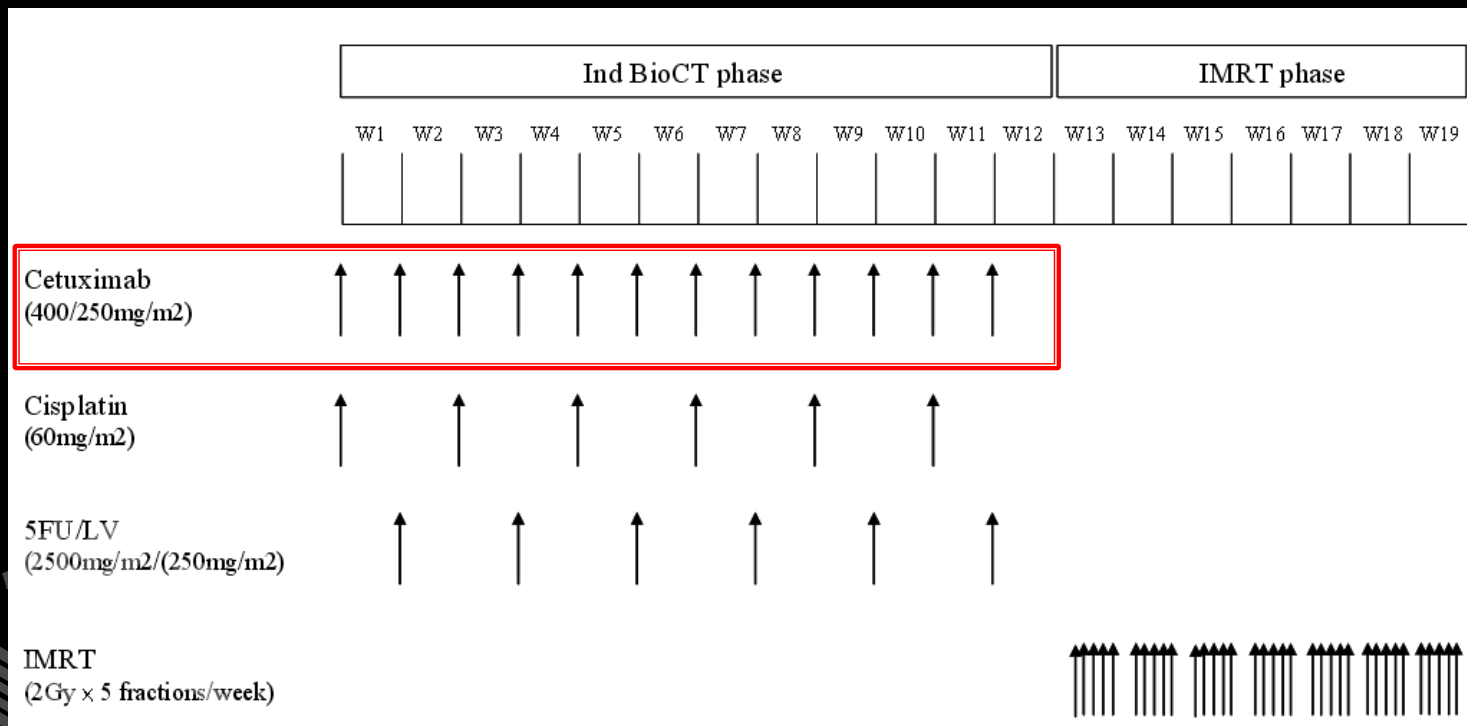
9(20%) Gr 1-2 hemorrhage

Lancet Oncol 2012;13:172-80

Induction BioChemotherapy and IMRT for Locally Advanced NPC

Department of Radiation Oncology
Taichung Veterans General Hospital

2013 2nd ESTRO Forum, Abstract PO-0666



PFL-based (33) or others (2)

TNM Staging (n=35)

Parameters	No.	%
T-stage		
T1	8	22.9
T2	13	37.1
T3	7	20.0
T4	7	20.0
N-stage		
N0	1	2.9
N1	2	5.7
N2	18	51.4
N3	14	40.0
Overall stage		
III	15	42.9
IV	20	57.1

Acute Reaction of NeoCT-Erbitux (n=35)

Grade	0	I	II	III	IV	n (%)
WBC	10 (29)	16 (46)	6 (17)	3 (9)	0	
Hb	14 (40)	12 (34)	7 (20)	1 (3)	1 (3)	
Platelet	30 (86)	2 (6)	2 (6)	1 (3)	0	
Mucositis	16 (46)	10 (29)	9 (26)	0	0	
Vomiting	18 (51)	12 (34)	5 (14)	0	0	
Weight loss	10 (29)	15 (43)	10 (29)	0	0	
Skin rash	2 (6)	16 (46)	17 (49)	0		

Paronychia = 20/35 (57.1%)

Dry skin, fissure = 15/35 (42.9%)

Hypomagnesia = 11/35 (31.4%)

Acute Reaction during IMRT (n=35)

Grade	0	I	II	III	IV	n (%)
WBC	11 (44)	16 (46)	8 (23)	0	0	
Hb	11 (44)	16 (46)	6 (17)	2 (6)	0	
Platelet	35 (100)	0	0	0	0	
Mucositis	1 (3)	11 (44)	17 (49)	6 (17)	0	↓
Dermatitis	1 (3)	20 (57)	13 (37)	1 (3)	0	
Vomiting	33 (94)	2 (6)	0	0	0	
Weight loss	5 (14)	17 (49)	13 (37)	0	0	

Response to NeoCT + Erbitux

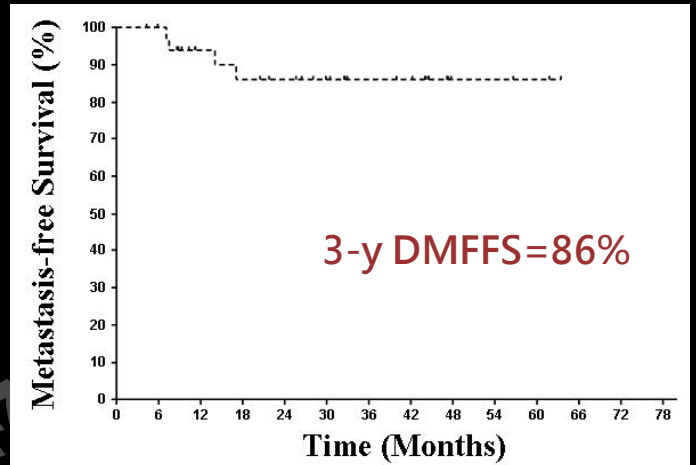
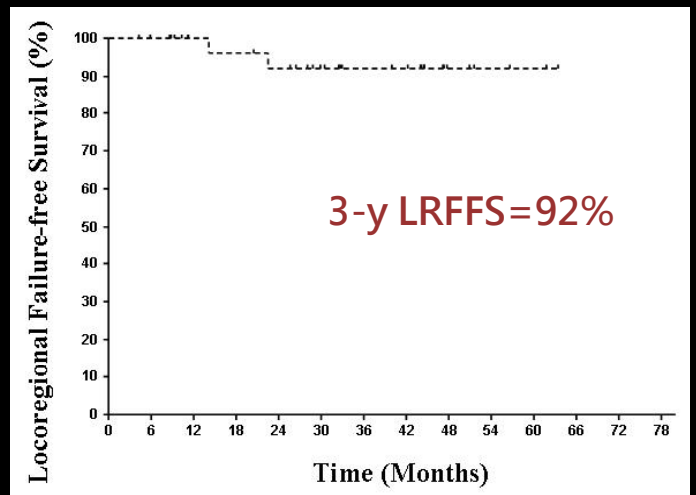
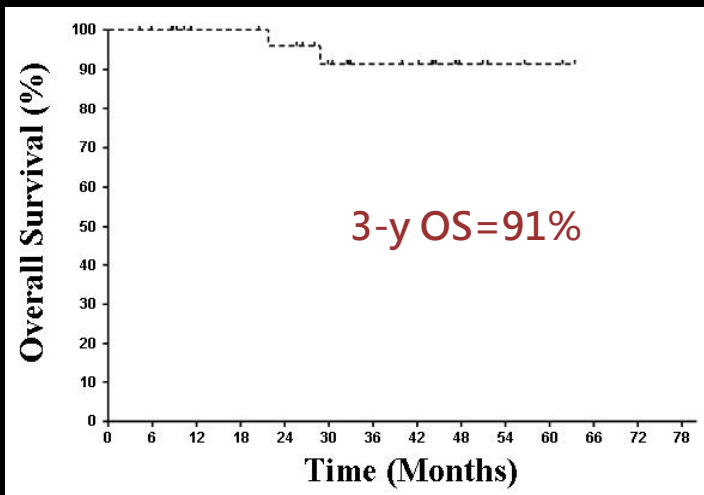
Clinical CR/PR = 17/18

CR = 48.6%; RR=100%

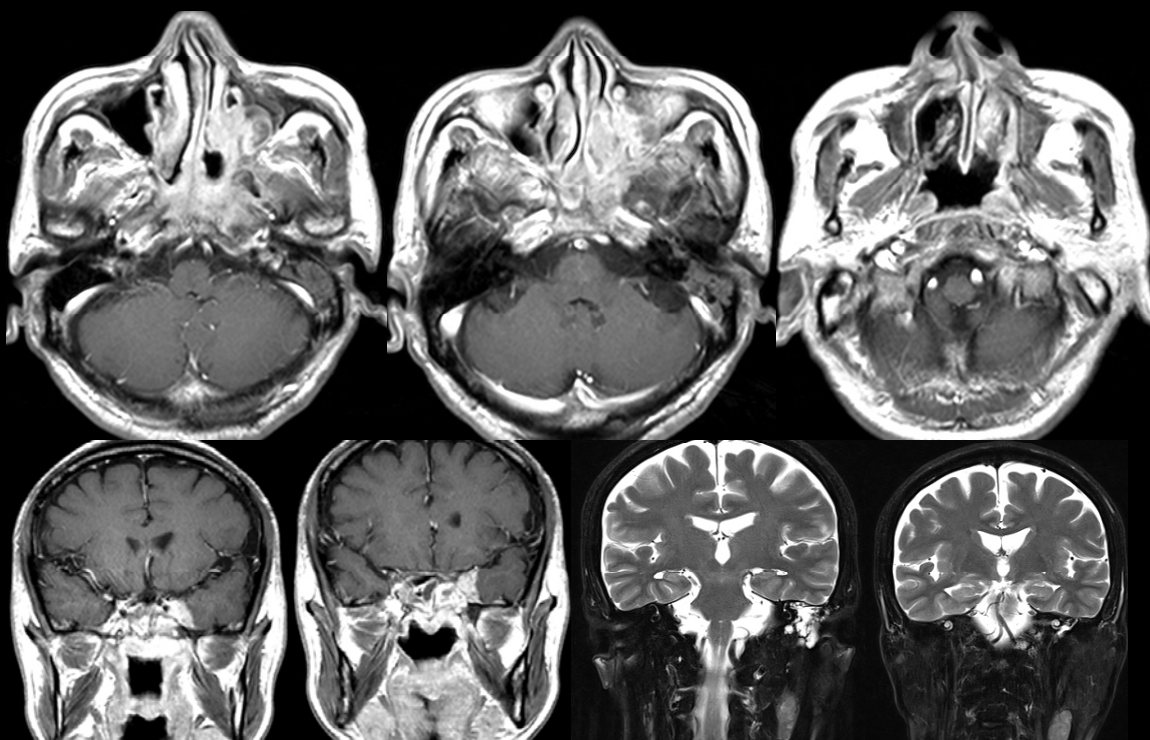
Re-Bx (-) before IMRT = 18/33 (54.5%)

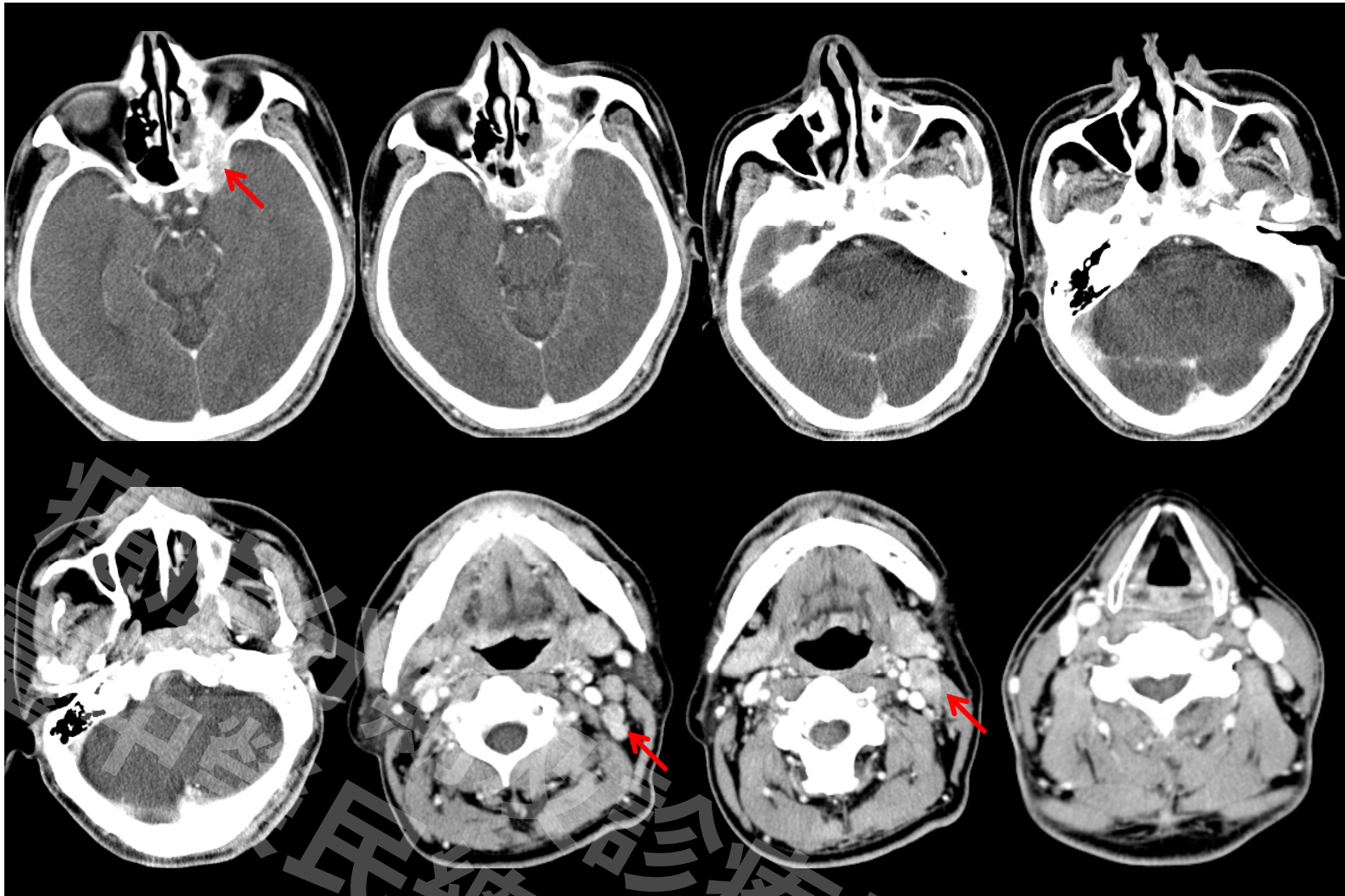
Failure pattern (n=35)

Local	1
Neck	1
Distant metastasis	4
axillary nodes	1
lung	1
liver	1
liver + lung	1



52/Male, Lt facial numbness, diplopia, bloody nasal discharge 4 m
 Neur OPD → MRA → NP tumor → ENT FESS: NKSCC, cT4N1M0





P-FL +
Erbitux
X 6 wk

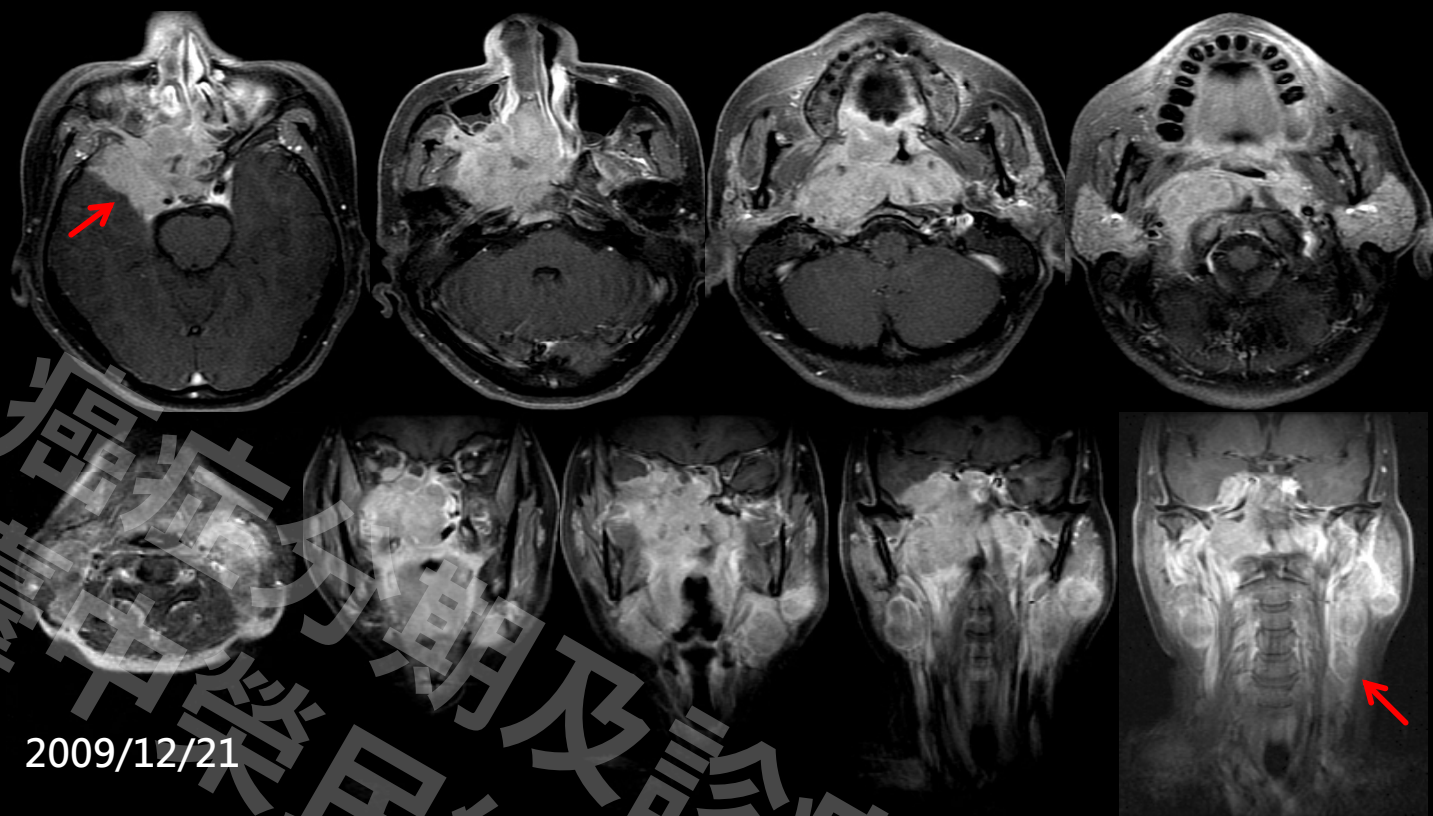


WBC 3600
Hb 12.1
Platelet 164k

Gr 1 skin rash

40/female, neck mass + epistaxis 6 m → ENT (2009/12/18)

Bx: NKSCC cT4N3a+3bM0



2009/12/21

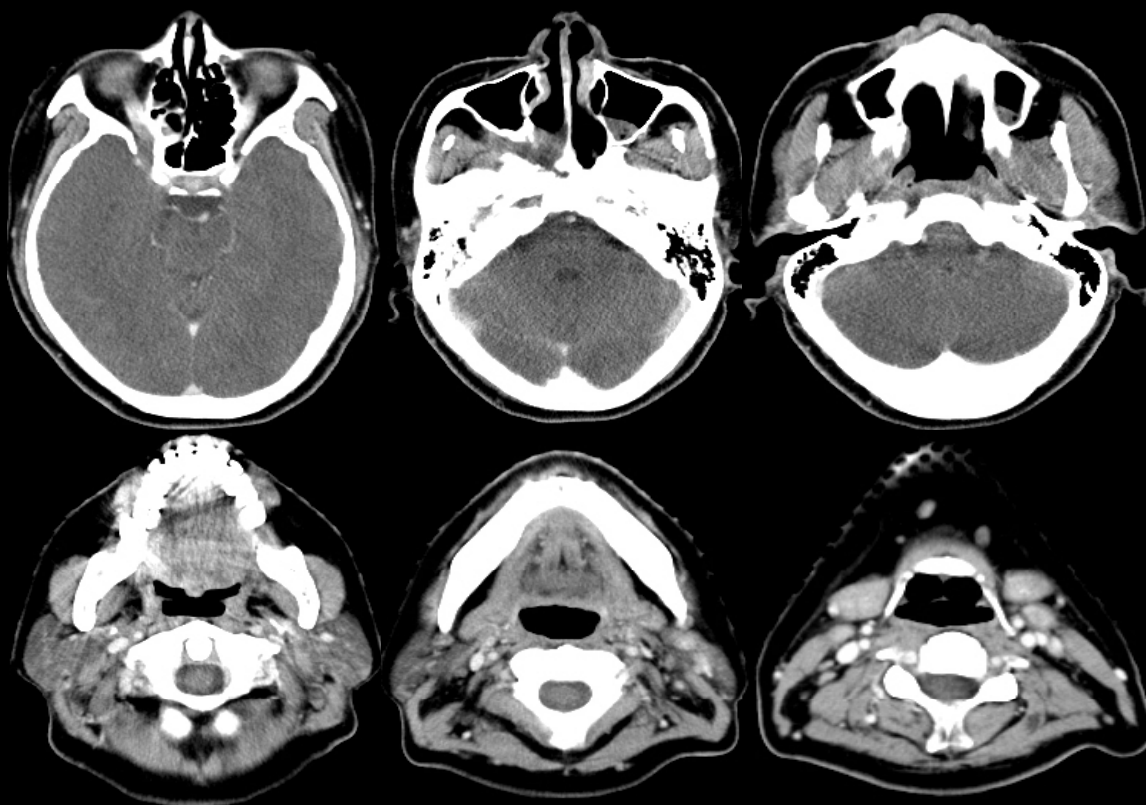
weekly (P-FL-G + Erbitux) x 12 (2009/12/25-2010/03/10) → CR

(2010/03/18) Bx: No tumor

Gr 1 dermatitis

IMRT 74 Gy/37fr/53d

Gr 3 mucositis



2010/03/10
CT Sim

Summary of Target therapy in NPC

Treatment	Erb-CCRT (P30qw) PWH	Erb-CCRT (P80q3w) SYSCC	Erb-IndCT + IMRT TC-VGH	Erb-CRT FDUCC	Avs-CCRT + Avs-PF RTOG0615
WHO type I/II-III	0/100	0/100	0/100	0/100	9/91
Stage II/III/IV	0/67/33	0/71/29	0/43/57	12/33/55	11/55/34
Gr ³ / ₄ Mucositis	87	51	17	84.8	77
Gr ³ / ₄ Dermatitis	20	9	3	18.2	14
Gr ³ / ₄ Leucopenia	10	6	9	0	36
PR/CR rate	13/83	18/80	6/94	12/88	-
LR control	93(2-y)	90(2-y)	92(3-y)	-	84(2-y)
MFS	93(2-y)	89(2-y)	86(3-y)	84(3-y)	91(2-y)
OS	91(2-y)	90(2-y)	91(3-y)	90(3-y)	91(2-y)
PFS	87(2-y)	89(2-y)	78(3-y)	71(3-y)	74(2-y)

Thank you for your attention